

Enterprise RAG · Week 2 — Portal Map

Course-portal map

This is a skeleton outline only. The full worked narrative, analogies, and derivations are delivered live in class and as the in-person printed handout.

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What you should be able to do

- Write the softmax equation from memory, and explain in one breath what the exponential does (makes everything positive and amplifies gaps) and what the denominator does (restores the conservation law, so the mass sums to one).
- Explain the role of temperature: show what happens to the distribution as $T \rightarrow 0$ (it snaps onto the argmax, a hard decision) and as $T \rightarrow \infty$ (it relaxes to the uniform distribution, a shrug), and connect both limits to the same exponential.
- Motivate surprise as $-\log p$ from first principles: why the inverse $1/p$ captures the intuition but starts at the wrong place, and why the logarithm is the natural repair that sends certainty to zero surprise.
- Give the rigorous reason the logarithm is forced, not chosen: that the likelihood is a product, that maximising a product of a thousand probabilities underflows any floating-point number ever built, and that $\log$ is the unique (up to scale) continuous map turning products into sums — so additivity and computability are the same fact, and they hand us the negative log-likelihood.
- Explain attention as an approximate, differentiable dictionary lookup: given a query that matches no key exactly, distribute a unit of attention over the nearby keys by proximity, and read off a blended value.
- State the scaled dot-product attention equation, identify which part is the softmax, which is the proximity score, and which is the retrieved meaning, and explain why the worked example must be a polysemous word like bank.
- Explain why machines must speak "machineese": that because they learn by gradient descent, meaning must live on a smooth, continuous manifold where infinitesimal moves are legal — something the discrete lexicon can never offer.
- Describe the bridge from per-token contextual meanings to a single chunk embedding: that attention yields one vector per word, and that pooling (mean-pooling, usually) collapses them into the one vector we actually index.
- Explain masked language modelling as the "guess the missing word" game, and argue why a machine that consistently wins it must have built genuine contextual understanding — and why its score is exactly the negative log-likelihood you built this morning.
- Explain anisotropy and the cure: that attraction alone collapses a space into a cone, that contrastive learning adds the missing repulsive force, and how to measure the improvement with within-class versus between-class cosine histograms.
- Read search as motion on a curved surface — the information-geometry view — and explain why a single shared space for text and images (CLIP) is what lets a sentence retrieve a photograph.

Part 1 — Softmax, temperature (softmax), surprise, negative log-likelihood, entropy, cross-entropy

Softmax

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

Temperature (softmax)

$$\text{softmax}_T(z)_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}.$$

Core formulae

$$\text{softmax}(z)_i = \frac{e^{z_i - \max_j z_j}}{\sum_k e^{z_k - \max_j z_j}}.$$

Surprise

$$S(p) = \log \frac{1}{p} = -\log p.$$

Negative log-likelihood

$$S_{\text{total}} = \sum_k -\log p_k = -\sum_k \log p_k.$$

Entropy, cross-entropy, perplexity

$$H(p) = -\sum_i p_i \log p_i = \mathbb{E}_p[-\log p].$$

$$H(p, q) = -\sum_i p_i \log q_i,$$

$$\text{PPL} = 2^H.$$

Core formulae

$$\frac{\partial \mathcal{L}}{\partial z} = p - y.$$

Maximum likelihood estimation

$$L = \prod_{k=1}^N p_k.$$

$$\log L = \log \prod_k p_k = \sum_k \log p_k.$$

Part 2 — Tokenization, manifold, self-attention, query, key, value, pooling, masked language modelling

Tokenization

Manifold

Self-attention

$$\text{ctx_meaning}(\text{bank}) = \sum_i s_i v_i,$$

$$s_i = \text{softmax}\left(\frac{\langle k_i, q \rangle}{\sqrt{d}}\right).$$

Query, key, value

Pooling

Core formulae

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

$$p(y_1, y_2, \dots, y_T) = \prod_{t=1}^T p(y_t \mid y_{<t}, x),$$

Masked language modelling

Part 3 — Anisotropy, contrastive learning

Anisotropy, contrastive learning

Core formulae

$$\mathcal{L}_{\text{triplet}} = \max(0, \|a - p\|^2 - \|a - n\|^2 + m).$$

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(a, p)/\tau)}{\sum_j \exp(\text{sim}(a, x_j)/\tau)}.$$

Core formulae

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|},$$

Core formulae

$$D_{\text{KL}}(p \parallel q) = \sum_i p_i \log \frac{p_i}{q_i} = H(p, q) - H(p),$$

Part 4 — Core formulae and terms

Core formulae

$$\text{Var}\left(\sum_{i=1}^d q_i k_i\right) = \sum_{i=1}^d \text{Var}(q_i k_i) = \sum_{i=1}^d \mathbb{E}[q_i^2] \mathbb{E}[k_i^2] = \sum_{i=1}^d 1 \cdot 1 = d.$$

Core formulae

$$\mathcal{L} = -\sum_i p_i \log p_i - \alpha \left(\sum_i p_i - 1 \right) - \beta \left(\sum_i p_i E_i - U \right).$$

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\log p_i - 1 - \alpha - \beta E_i = 0 \implies p_i = e^{-1-\alpha} e^{-\beta E_i}.$$

$$p_i = \frac{e^{-E_i/kT}}{\sum_j e^{-E_j/kT}}.$$

Core formulae

$$E(q) = -\text{lse}(\beta, K^\top q) + \frac{1}{2} \|q\|^2 + \text{const},$$

$$q^{\text{new}} = V \text{softmax}(\beta K^\top q),$$

Required reading

- Vaswani et al., Attention Is All You Need
- Devlin et al., BERT: Pre-training of Deep Bidirectional Transformers
- Reimers & Gurevych, Sentence-BERT
- Radford et al., Learning Transferable Visual Models from Natural Language
- Ethayarajh, How Contextual Are Contextualized Word Representations?

Optional further reading

- Shannon, A Mathematical Theory of Communication
- Jaynes, Information Theory and Statistical Mechanics
- Ramsauer et al., Hopfield Networks Is All You Need
- Cover & Thomas, Elements of Information Theory
- Su et al., Whitening Sentence Representations